Differential Equation

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Outline of Presentation

1. Linear differential equation of higher order
   - Homogeneous Equation with Constant Coefficients
   - Theory of Solutions of Linear Differential Equations
   - Reduction Order
   - Linear Non-Homogeneous Equation with Constant Coefficients
   - Method of Undetermined Coefficients
   - Method of Variation of Parameters
2. Linear Differential Operator Method (D-Operator)

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Linear differential equation of higher order

A general nth order linear differential equation has the form:

$$a_n(x)y^n + a_{n-1}(x)y^{n-1} + \cdots + a_2(x)y'' + a_1(x)y' + a_0(x)y = f(x) \quad (1)$$

This equation (16) is homogeneous if $f(x) = 0$ and non-homogeneous if $f(x) \neq 0$.
The solution of the homogeneous differential equation is called the complimentary function (conventionally y_c) and the solution of the non-homogeneous part is known as the particular integral (conventionally y_p).

Homogeneous Equation with Constant Coefficients

A general homogeneous differential equation with constant coefficients is:

$$a_n y^n + a_{n-1} y^{n-1} + \cdots + a_2 y'' + a_1 y' + a_0 y = 0 \qquad (2)$$

We start problem solving with second order differential equation of the form:

$$a_2 y'' + a_1 y' + a_0 y = 0 \qquad (3)$$

We consider a trial solution $y = e^{rx}$.We can write (18) as $ar^2 + br + c = 0$.Hence if r is root of this quadratic equation,often called the auxiliary or characteristic equation.

There are three cases to consider:

1. If the roots are real and distinct,that is $r = r_1, r = r_2$ and $r_1 \neq r_2$ then the general solution of the differential equation is:

$$y = c_1 e^{r_1 x} + c_2 e^{r_2 x} \tag{4}$$

2. If the roots are real and identical,that is $r = r_1, r = r_2$ and $r = r_1 = r_2$ then the general solution of the differential equation is:

$$y = c_1 e^{r_1 x} + c_2 x e^{r_2 x} \tag{5}$$

3. If the roots are not real,the roots are complex conjugates,that is $r = \alpha \pm i\beta$ general solution of the differential equation is:

$$y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x) \tag{6}$$

Example

1. Solve the differential equation $y'' + 2y' + y = 0$
   Solution
   $r^2 + 2r + 1 : r = -1$

$$y = c_1 e^{-x} + c_2 x e^{-x}$$

2. Solve the differential equation $y'' + 2y' + 5y = 0$
   Solution
   $r^2 + 2r + 5 : r = -1 \pm 2i$

$$y = e^{-x}(c_1 \cos 2x + c_2 \sin 2x)$$

3. Solve the differential equation $y'' + y' - 6y = 0$
   Solution
   $r^2 + r - 6 : r_1 = 2, r_2 = -3$

$$y = c_1 e^{-x} + c_2 x e^{-x}$$

Theory of Solutions of Linear Differential Equations

Consider a Homogeneous second-order linear differential equation

$$y'' + p(x)y' + q(x)y = 0 \tag{7}$$

Principle of Superposition

Let y_1 and y_2 be two solutions of the homogeneous linear equation (7) on the interval I. If c_1 and c_2 are constants, then the linear combination

$$y = c_1 y_1 + c_2 y_2 \tag{8}$$

is a solution of (7) on I.

Linear Independence of Two Functions and the Wronskian

Two functions defined on an open interval are called linearly independent provided that neither is a constant multiple of the other. Two functions are said to be linearly dependent if they are not linearly independent; that is, one is a constant multiple of the other.

Given two functions f and g , the Wronskian of f and g is the determinant : $\begin{bmatrix} f & g \\ f' & g' \end{bmatrix} = fg' - f'g$

Example

$$W(\cos x, \sin x) = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix} = \cos^2 x + \sin^2 x = 1 \neq 0$$

if the $W(f, g) \neq 0$ then f and g are linearly independent otherwise Linearly dependent.

Reduction Order

Consider the non-constant coefficient, second order differential equations of the form

$$y'' + p(x)y' + q(x)y = 0 \tag{9}$$

Solution of Reduction Order:

Consider

$$y'' + p(x)y' + q(x)y = 0 \tag{10}$$

Let

$$y_2(x) = v(x)y_1(x) \tag{11}$$

Substitute (11) into (10), using the derivatives:

$$y_2' = vy_1' + v'y_1 \qquad \Longrightarrow \; y_2'' = vy_1'' + 2v'y_1' + v''y_1 \tag{12}$$

We get

$$[vy_1'' + 2v'y_1' + v''y_1] + p[vy_1' + v'y_1] + qvy_1 = 0 \tag{13}$$

$$v[y_1'' + py_1' + qy_1] + v''y_1 + 2v'y_1' + pv'y_1 = 0 \tag{14}$$

But $v_1'' + py_1' + qy_1 = 0$ is a solution of (10).

$$y_1 v'' + (2y_1' + py_1)v' = 0 \tag{15}$$

If we write $u = v'$ and assume that y_1 never vanishes on I, then (11) yields:

$$u' + \left(\frac{2y_1'}{y_1} + p(x) \right) u = 0 \tag{16}$$

An integrating factor will be:

$$\mu = \exp \left\{ \int \left[2\frac{y_1'}{y_1} + p(x) \right] dx \right\} = \exp \left[\ln |y_1| + \int p(x) dx \right] \tag{17}$$

Thus

$$\mu = y_1^2 e^{\int p(x) dx} \tag{18}$$

We now integrate the equation in (10) to obtain:
$y_1^2 e^{\int p(x)dx} = c$
So

$$u = \frac{c}{y_1^2} e^{-\int p(x)dx}. \tag{19}$$

Another integration gives:

$$\frac{y_2}{y_1} = v = c \int \frac{e^{-\int p(x)dx}}{y_1^2} dx + k. \tag{20}$$

With the particular choices $c = 1$ and $k = 0$, we get:

$$y_2 = y_1 \int \frac{e^{-\int p(x)dx}}{y_1^2} dx \tag{21}$$

Example

Find the general solution to $2x^2y'' + xy' - 3y = 0$

Solution:

$$
\begin{aligned}
& 2x^2y'' + xy' - 3y = 0 \\
\implies \quad & y'' + \frac{1}{2x}y' + \frac{3}{2x^2} = 0 \quad \text{where } p(x) = \frac{1}{2x} \\
& y_2 = y_1 \int \frac{e^{-\int p(x)dx}}{d}x \\
\implies \quad & y_2 = x^{-1} \int \frac{e^{-\int \frac{1}{2x}dx}}{(x^{-1})^2} = x^{-1} \int \frac{x^{-\frac{1}{2}}}{x^{-2}}dx
\end{aligned}
$$

$$\Longrightarrow \quad y_2 = x^{-1}\int x^{\frac{3}{2}}dx = x^{-1}\left(\frac{2}{5}x^{\frac{5}{2}} + c\right)$$

let $\qquad c = 0 \quad y_2 = \frac{2}{5}x^{\frac{3}{2}}$

The general solution is: $y = c_1x^{-1} + c^2x^{\frac{3}{2}}$.

Linear Non-Homogeneous Equation with Constant Coefficients

A general linear n-n-homogeneous coefficient equation is:

$$a_n = \frac{d^n y}{dx^n} + a_{n-1}\frac{d^{n-1}}{dx^{n-1}} + \cdots + a_1\frac{dy}{dx} + a_0 y = f(x) \qquad (22)$$

$f(x)$ is a given function of x, and a_0, a_1, $\cdots$, a_{n-1}, a_n also given constants.

Method of Undetermined Coefficients

1. Assume a trial solution function $y_p(x)$, a family of functions that is guaranteed to include a correct particular solution;
2. Find the image of the trial solution whose form depends on $f(x)$ of (22),
3. $y_p(x)$ is then substituted into (22) and the constants take values which satisfy (22) completely.

Example

Find the particular integral of $y'' - 3y' + 2y = e^{2x}$

Solution:

The auxiliary equation is:

$r^2 - 3r + 2 = 0, \implies (r-2)(r-1) = 0, \quad \therefore r = 2, 1$

Hence $f(x) = e^{2x}$ that is $b = 2$, a root.

The trial function is: $y_p(x) = Axe^{2x}$

Substituting into the differential equation:

$$y_p(x) = Axe^{2x},\ y_p'(x) = Ae^{2x} + 2Axe^{2x}$$

$$y_p''(x) = 2Ae^{2x} + 2Ae^{2x} + 4Axe^{2x}$$

$$\implies 4Ae^{2x} + 4Axe^{2x} - 3Ae^{2x} - 6Axe^{2x} + 2Axe^{2x} = e^{2x}$$

$$\implies Ae^{2x} = e^{2x} \quad \implies A = 1$$

$$\therefore y_p(x) = xe^{2x}$$

General solution: $y = y_c + y_p = c_1e^{2x} + c_2e^{2x} + xe^{2x}$

Method of Variation of Parameters

In this method, the particular integral is expressed as the modification of the complementary solution. This is done by making of the coefficient of the complementary solution change from the constants to variables.
Thus we have

$$y_p(x) = c_1(x)\, y_1(x) + c_2(x)\, y_2(x)$$

where $y_1(x)$ and $y_2(x)$ are independent solutions of the homogeneous equations.

$$\begin{aligned} y\prime_p &= c\prime_1(x)\, y_1(x) + c\prime_2(x)\, y_2(x) \\ &+ c_1(x)\, y\prime_1(x) + c_2(x)\, y\prime_2(x) \end{aligned}$$

assume for simplicity that

$$c_1'(x)\,y_1(x) + c_2'(x)\,y_2(x) = 0 \tag{23}$$

so that

$$\begin{aligned} y_p' &= c_1'(x)\,y_1(x) + c_2'(x)\,y_2(x) \\ \Longrightarrow\ y_p''(x) &= c_1'(x)\,y_1(x) + c_2'(x)\,y_2(x) \\ &+ c_1(x)\,y_1''(x) + c_2(x)\,y_2''(x) \end{aligned}$$

$$\begin{aligned} y_p'' &= c_1(x)\,y_1''(x) + c_2(x)\,y_2''(x) + c_1'(x)\,y_1'(x) + c_2'(x)\,y_2'(x) \\ a_2\,y_p' &= c_1(x)\,a_2\,y_1' + c_2(x)\,a_2\,y_2' \\ a_1\,y_p &= c_1(x)\,a_1\,y_1 + c_2(x)\,a_2\,y_2 \end{aligned}$$

$$
\begin{aligned}
y_p^0 + a_2\, y_p' + a_1\, y_p &= c_1(x)[y_1'' + a_2\, y_1' + a_1\, y_1] \\
&+ c_2(x)[y_2'' + a_2\, y_2' + a_2\, y_2] \\
&+ c_1'(x)\, y_1'(x) + c_2'(x)\, y_2'(x) \\
&= c_1'(x)\, y_1'(x) + c_2'(x)\, y_2'(x)
\end{aligned}
$$

since

$$\begin{aligned} y_2 + a_2\, y_2' + a_2 y_1 &= 0 \\ \Longrightarrow\ y_2' + a_2\, y_2' + a_1\, y_2 &= 0 \end{aligned}$$

$y_1, \quad y_2$ being solution of the homogeneous equation.

$$c_1'(x)\, y_1'(x) + c_2'(x)\, y_2'(x) = f(x) \tag{24}$$

Note that equations (23) and (24) give the system:

$$\begin{pmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{pmatrix} \begin{pmatrix} c_1'(x) \\ c_2'(x) \end{pmatrix} \begin{pmatrix} 0 \\ f(x) \end{pmatrix}$$

.

Solving by means of rule, we have

$$c\prime_1(x) = \frac{\begin{vmatrix} 0 & y_2(x) \\ f(x) & y\prime_2(x) \end{vmatrix}}{w(y_1(x,\, y_2(x)))} = \frac{-y_2(x)\, f(x)}{w(y_1,\, y_2)}$$

$$c\prime_2(x) = \frac{\begin{vmatrix} y_1 & 0 \\ y_2 & f(x) \end{vmatrix}}{w(y_1(x,\, y_2(x)))} = \frac{y_1(x)\, f(x)}{w(y_1,\, y_2)}$$

Therefore

$$c_1(x) = \int \frac{y_2(x)\, f(x)}{w(y_1(x),\, y_2(x))}\, dx + k_1$$

$$c_2(x) = \int \frac{y_1(x)\, f(x)}{w(y_1(x),\, y_2(x))}\, dx + k_2$$

Example:

Solve the IVP using the method of variation of parameters.

$$y'' + 3y' + 2y = \sin x$$
$$y(0) = 0,\ y'(0) = 1$$

Solution

$$\begin{aligned} \text{put } y(x) &= c_1(x)\, y_1(x) + c_2(x)\, y_2(x) \\ \text{have } y_1(x) &= e^{-2x}, \qquad y_2(x) = e^{-x} \\ \text{Thus } y(x) &= c_1(x)\, e^{-2x} + c_2(x)\, e^{-x} \end{aligned}$$

:

Solution Cont'd

Applying the method of variation of parameters from the first principles, we have:

$$\begin{aligned} y\prime(x) &= c\prime_1(x)\, e^{-2x} + c_2'(x)\, e^{-x} \\ &- 2c_1(x)\, e^{-2x} - c_2(x)\, e^{-x} \end{aligned}$$

We assume:

$$c\prime_1(x)\, e^{-2x} + c\prime_2(x)\, e^{-x} = 0 \tag{25}$$

$$\begin{aligned} \implies y\prime(x) &= -2c_1(x)\, e^{-2x} - c_2(x)\, e^{-x} \\ \implies y\prime\prime(x) &= -2c\prime_1(x)\, e^{-2x} - c\prime_2(x)\, e^{-x} \\ &+ 4c_1(x)\, e^{-2x} - c_2(x)\, e^{-x} \\ \implies 3y\prime(x) &= 3(-2c_1(x)\, e^{-2x} - c_2(x)\, e^{-x}) \\ 2y(x) &= 2(c_1(x)\, e^{-2x} - c_2(x)\, e^{-x}) \\ y\prime\prime(x) + 3y\prime(x) + 2y(x) &= -2c_1(x)\, e^{-2x} - c_2(x)\, e^{-x} \end{aligned}$$

Therefore

$$-2c_1(x)\,e^{-2x} - c_2(x)\,e^{-x} = \sin x \tag{26}$$

Equations (25) and (26) in matrix notation gives:

$$\begin{pmatrix} e^{-2x} & e^{-x} \\ -2e^{-2x} & -e^{-x} \end{pmatrix} \begin{pmatrix} c\prime_1(x) \\ c\prime_2(x) \end{pmatrix} = \begin{pmatrix} 0 \\ \sin x \end{pmatrix}$$

$$\implies c\prime_1(x) = \frac{\begin{pmatrix} 0 & e^{-x} \\ \sin x & -e^{-x} \end{pmatrix}}{e^{-3x}} = \frac{-e^{-x}\sin x}{e^{-3x}} = -e^{2x}\sin x$$

$$\implies c_2'(x) = \frac{\begin{pmatrix} e^{-2x} & 0 \\ -2e^{-2x} & \sin x \end{pmatrix}}{e^{-3x}} = \frac{e^{-2x}\sin x}{e^{-3x}} = e^{x}\sin x$$

$$\begin{aligned} c_1(x) &= -\int^x e^{2t}\sin t dt + A_1 \\ &= te^{2x}\cos x - 2\int \cos te^{2t} dt + A_1 \end{aligned}$$

$$\begin{aligned} &= e^{2x}\cos x - 2\left[e^{2x}\sin x - 2\int e^{2t}\sin t dt \right. \\ \implies -5\int^x e^{2t}\sin t dt &= e^{2x}\cos x - 2e^{2x}\sin x + A_1 \\ c_1(x) &= \frac{1}{5}e^{2x}[\cos x - 2e^{2x}\sin x] + A_1 \end{aligned}$$

Similarly

$$\begin{aligned} c_2(x) &= \int^x e^t \sin t dt + A_2 \\ &= -e^x \cos x + \int^x e^t \cos t dt + A_2 \\ &= -e^x \cos x + \left[e^x \sin x - \int e^t \sin t dt \right] + A_2 \\ \implies 2 \int^x e^t \sin t dt &= e^x [\sin x - \cos x] + A_2 \\ \implies c_2(x) &= \frac{1}{2} e^x [\sin x - \cos x] + A_2 \end{aligned}$$

Now the fundamental functions are
$y_1(x) = e^{-2x}$ and $y_2(x) = e^{-x}$
Therefore

$$\begin{aligned} y &= c_1(x)e^{-2x} + c_2(x)e^{-x} + A_1e^{-2x} + A_2e^{-x} \\ &= \frac{1}{5}[\cos x - 2\sin x] + \frac{1}{2}[\sin x - \cos x] \\ &= \frac{2}{10}\cos x - \frac{5}{10}\cos x - \frac{4}{10}\sin x + \frac{5}{10}\sin x \\ &= \frac{1}{10}[\sin x - 3\cos x] \end{aligned}$$

Thus the general solution is given by

$$y = A_1 e^{-2x} + A_2 e^{-x} + \frac{1}{10}[\sin x - 3\cos x]$$

Example

Find the general solution of the differential equation

$$y'' + y = \sec x \qquad 0 < x < \frac{\pi}{2}$$

Solution:

The two general solution of the homogeneous equation are:

$$y_1 = \cos x \text{ and } y_2 = \sin x$$

Then

$$y_p = u_1 \cos x + u_2 \sin x$$
$$\implies y_p' = u_1' \cos x - u_1 \sin x + u_2' \sin x + u_2 \cos x$$

$$u_1' \cos x + u_2' \sin x \tag{27}$$

$$y_p' = -u_1(x) \sin x + u_2(x) \cos x$$
$$y_p'' = -u_1' \sin x - u_1 \cos x + u_2' \cos x - u_2 \sin x.$$

Then the differential equation becomes:

$$-u_1' \sin x - u_1 \cos x + u_2' \cos x - u_2 \sin x + u_1 \cos x + u_2 \sin x = \sec x$$

$$\implies -u_1' \sin x + u_2' \cos x = \sec x \tag{28}$$

Using (27) and (28), we have:

$$u_1' \cos x + u_2' = 0$$
$$-u_1' \sin x + u_2' \cos x = \sec x$$

The Cramer's Rule for the solution of a linear system of equation yields:

$$u_1' = \frac{\begin{vmatrix} 0 & \sin x \\ \sec x & \cos x \end{vmatrix}}{\begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}} \qquad u_2' = \frac{\begin{vmatrix} \cos x & 0 \\ -\sin x & \sec x \end{vmatrix}}{\begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}}$$

$$\implies u_1' = \tan x, \qquad \implies u_2' = 1$$

$$\implies u_1 = \ln \cos x, \qquad \implies u_2 = x$$

Therefore, $y_p = (\ln \cos x)\cos x + x \sin x$
General solution:

$$y(x) = c_1 \cos x + c_2 \sin x + (\ln \cos x)\cos x + x \sin x$$

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Example

1. $D(x^2) = 2x$
2. $D^2(e^{3x}) = 9e^{3x}$
3. $D^2(x^3 - 4x^2 + 1) = 6x - 8$

Definition

A differential operator of order n can be expressed as

$$D$$

THANK YOU